6.2 Building the Map from the Rationals Below

Section 6.1 gave every element x of an ordered field F a record, the set $\mathbb{Q}_F(x)$ of rationals below it, and showed that in an Archimedean field the record determines the element (Lemma 6.1.3). This section turns the record into a map. Given an Archimedean field F and a complete field L, we send $x \in F$ to the element of L with the same record. The result is the main theorem of the chapter in its one-sided form: there is exactly one order-preserving field map $F \to L$. Section 6.3 applies it twice, in both directions, to compare two complete ordered fields.

Two fields are in play throughout, so every proof here keeps ι_F and ι_L visible. Only the two lemmas, each about one field, write q for $\iota_F(q)$.

6.2.1 What the map must be

Suppose $\varphi\colon F \to L$ is an order-preserving field map (Definition 6.1.1). By Proposition 6.1.1 it fixes the rationals, $\varphi(\iota_F(q)) = \iota_L(q)$, and it preserves and reflects order. So for $q \in \mathbb{Q}$ and $x \in F$,

$$\iota_F(q) < x \iff \varphi(\iota_F(q)) < \varphi(x) \iff \iota_L(q) < \varphi(x).$$

In the notation of Section 6.1: $\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x)$. *The image of x has exactly the rationals below it that x has.*

That already shows there can be at most one such map when L is Archimedean: two candidates for $\varphi(x)$ would have the same rationals below them, and rationals separate. For existence, we need an element of L with a prescribed set of rationals below it. When L is complete there is a natural candidate: the least element of L that lies above all the copies $\iota_L(q)$, $q \in \mathbb{Q}_F(x)$, that is, their supremum.

The check that this candidate respects sums and products goes through the record, not through suprema. For that we need to know how the rationals below $a + b$ and below ab are made from those below a and b.

6.2.2 Sums and products, measured by rationals

A rational q lies below $a + b$ exactly when it splits into a rational below a and a rational below b.

Lemma 6.2.1. Rationals below a Sum

Let F be an Archimedean ordered field, $a, b \in F$ and $q \in \mathbb{Q}$. Then $\iota_F(q) < a + b$ if and only if there is $r \in \mathbb{Q}$ with

$$\iota_F(r) < a \qquad \text{and} \qquad \iota_F(q - r) < b.$$

In other words, $q \in \mathbb{Q}_F(a + b) \iff \exists r \in \mathbb{Q}\ \big(r \in \mathbb{Q}_F(a) \wedge q - r \in \mathbb{Q}_F(b)\big)$.

Idea. One direction is addition of inequalities. For the other, split q as $r + (q - r)$ with r a rational just below a: we need r in the gap between $q - b$ and a, which is non-empty exactly because $q < a + b$, and density supplies it.

Proof. We write q for $\iota_F(q)$; since ι_F preserves sums, it preserves differences, so $\iota_F(q-r) = q-r$ in F.

($\Leftarrow$) If $r < a$ and $q - r < b$, then $q = r + (q-r) < a + b$ by Proposition 2.3.5 (b).

($\Rightarrow$) Let $q < a + b$. Then $q - b < a$ (Proposition 2.3.5 (a)). By Theorem 4.2.5 there is $r \in \mathbb{Q}$ with $q - b < r < a$. Then $r < a$, and $q - r < b$ by Proposition 2.3.5 (a) again.

■

For products the same idea works for positive elements and positive rationals, with q/r in place of $q - r$.

Lemma 6.2.2. Rationals below a Product

Let F be an Archimedean ordered field, $a, b \in F$ with $a > 0$ and $b > 0$, and $q \in \mathbb{Q}$ with $q > 0$. Then $\iota_F(q) < ab$ if and only if there is $r \in \mathbb{Q}$ with

$$r > 0, \qquad \iota_F(r) < a \qquad \text{and} \qquad \iota_F(q/r) < b.$$

For $q \leq 0$, $\iota_F(q) < ab$ holds always.

In words: below a product of positives, a positive rational splits as a product of a rational below each factor. Exercise B1 asks for the proof; the fold below has it.

↻ **Second reading** · *The proof of the product lemma (try Exercise B1 first)*

Idea. As for sums, with division in place of subtraction: split q as $r \cdot (q/r)$ with r a rational in the gap between q/b and a, a gap of positive elements.

Proof. We write q for $\iota_F(q)$; since ι_F preserves products and $\iota_F(r) \neq 0$ for $r \neq 0$, it preserves quotients, so $\iota_F(q/r) = q/r$ in F.

If $q \leq 0$, then $q \leq 0 < ab$ by (O4). Let $q > 0$.

($\Leftarrow$) Let $r > 0$ with $r < a$ and $q/r < b$. Since $q/r > 0$ (Proposition 2.5.2 (a), (O4)), Proposition 2.3.5 (c) gives $q = r \cdot (q/r) < a \cdot (q/r)$; since $a > 0$, it gives $a \cdot (q/r) < ab$. So $q < ab$.

($\Rightarrow$) Let $q < ab$. Multiplying by $b^{-1} > 0$ (Proposition 2.5.2 (a), Proposition 2.3.5 (c)) gives $q/b < a$, and $q/b > 0$ by (O4). By Theorem 4.2.5 there is $r \in \mathbb{Q}$ with $q/b < r < a$. Then $r > q/b > 0$, so $r > 0$ in $\mathbb{Q}$, since ι_F reflects order. Multiplying $q/b < r$ by $b > 0$ gives $q < rb$, and multiplying by $r^{-1} > 0$ gives $q/r < b$.

■

Quick check. In $F = \mathbb{Q}$, with $a := 3/4$, $b := 1/2$ and $q := 1$, find a rational r as in Lemma 6.2.1. Which rationals r work?

Solution. We need $r < 3/4$ and $1 - r < 1/2$, that is, $1/2 < r < 3/4$. For example $r = 2/3$: then $1 - 2/3 = 1/3 < 1/2$. Exactly the rationals of the open gap between $q - b = 1/2$ and $a = 3/4$ work, as the proof predicts.

6.2.3 The map into a complete field

Here is the theorem. Its proof is the first key proof of the chapter.

Theorem 6.2.3. Embedding an Archimedean Field in a Complete One

Let F be an Archimedean ordered field and L a complete ordered field. Then there is **exactly one** order-preserving field map $\varphi\colon F \to L$. It is given by

$$\varphi(x) := \sup_L \{\iota_L(q) : q \in \mathbb{Q}_F(x)\} \qquad (x \in F),$$

the supremum taken in L, and it satisfies

$$\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x) \qquad \text{for every } x \in F. \tag{$\star$}$$

Try it first (ten minutes): if φ exists and fixes $\mathbb{Q}$, which element of L must $\varphi(x)$ be?

Idea. The natural first attempt defines φ on the copy of $\mathbb{Q}$, by $\varphi(\iota_F(q)) := \iota_L(q)$, and then tries to "extend". It stalls at once: an element $x \in F$ that is not a rational has no formula, and nothing in F names it. What we do know, from the previous subsection, is its record: $\varphi(x)$ must have exactly the rationals $\mathbb{Q}_F(x)$ below it. So we **construct a point as a supremum**: $\varphi(x)$ is the least upper bound in L of the copies of those rationals, and $(\star)$ says the construction has the right record. After that, sums, products and uniqueness are checked the same way: compute the record of both sides (for sums and products by the two lemmas) and let rationals separate in L. Monotonicity needs only one rational between x and y.

Step roadmap: ① $\varphi(x)$ exists; ② $(\star)$; ③ φ fixes the rationals; ④ φ is strictly increasing; ⑤ sums; ⑥ products and the unit; ⑦ uniqueness. *Before reading on: where is completeness of L used, and where the Archimedean property of F?*

Proof. By Theorem 6.1.2, L is Archimedean. So Theorem 4.2.5, Lemma 6.1.3, Lemma 6.2.1 and Lemma 6.2.2 apply in L as well as in F. For $x \in F$ write $A_x := \{\iota_L(q) : q \in \mathbb{Q}_F(x)\} \subseteq L$.

Step 1: $\varphi(x)$ is defined. Let $x \in F$. Since F is Archimedean, there is $n \in \mathbb{N}$ with $-x < n \cdot 1_F$, so $\iota_F(-n) = -(n \cdot 1_F) < x$ (Proposition 2.3.5 (a)), and $-n \in \mathbb{Q}_F(x)$: the set A_x is non-empty. Applied to x, the Archimedean property gives $m \in \mathbb{N}$ with $x < m \cdot 1_F = \iota_F(m)$. For $q \in \mathbb{Q}_F(x)$, $\iota_F(q) < x < \iota_F(m)$, so $q < m$ because ι_F reflects order, and $\iota_L(q) < \iota_L(m)$. So $\iota_L(m)$ is an upper bound of A_x. Since L is complete, A_x has a supremum in L, and it is

unique; $\varphi(x)$ is that supremum. So $\varphi\colon F \to L$ is a function.

Step 2: $(\star)$**.** Let $x \in F$ and $q \in \mathbb{Q}$.

$(\subseteq)$ Let $q \in \mathbb{Q}_F(x)$. By density in F there is $r \in \mathbb{Q}$ with $\iota_F(q) < \iota_F(r) < x$. Then $r \in \mathbb{Q}_F(x)$, so $\iota_L(r) \in A_x$ and $\iota_L(r) \leq \varphi(x)$, since $\varphi(x)$ is an upper bound. Also $q < r$ (reflection in F), so $\iota_L(q) < \iota_L(r) \leq \varphi(x)$, and $q \in \mathbb{Q}_L(\varphi(x))$.

$(\supseteq)$ Let $\iota_L(q) < \varphi(x)$. Since $\varphi(x)$ is the **least** upper bound of A_x, $\iota_L(q)$ is not an upper bound, so there is $r \in \mathbb{Q}_F(x)$ with $\iota_L(q) < \iota_L(r)$ (totality). Then $q < r$ (reflection in L), so $\iota_F(q) < \iota_F(r) < x$, and $q \in \mathbb{Q}_F(x)$.

Step 3: φ **fixes the rationals.** Let $r \in \mathbb{Q}$. By $(\star)$ and the first example after Definition 6.1.2, used in F and in L,

$$\mathbb{Q}_L(\varphi(\iota_F(r))) = \mathbb{Q}_F(\iota_F(r)) = \{q \in \mathbb{Q} : q < r\} = \mathbb{Q}_L(\iota_L(r)).$$

By Lemma 6.1.3 in L, $\varphi(\iota_F(r)) = \iota_L(r)$. In particular $\varphi(0_F) = 0_L$ and $\varphi(1_F) = 1_L$.

Step 4: φ **is strictly increasing.** Let $x < y$ in F. By density in F there is $r \in \mathbb{Q}$ with $x < \iota_F(r) < y$. Then $r \notin \mathbb{Q}_F(x)$ (by (O1)), so $r \notin \mathbb{Q}_L(\varphi(x))$ by $(\star)$, and $\varphi(x) \leq \iota_L(r)$ by totality. And $r \in \mathbb{Q}_F(y) = \mathbb{Q}_L(\varphi(y))$, so $\iota_L(r) < \varphi(y)$. Hence $\varphi(x) \leq \iota_L(r) < \varphi(y)$.

Step 5: sums. Let $x, y \in F$ and $q \in \mathbb{Q}$. Then

$$\begin{aligned} q \in \mathbb{Q}_L(\varphi(x+y)) &\iff q \in \mathbb{Q}_F(x+y) \\ &\iff \exists r \in \mathbb{Q}\ (r \in \mathbb{Q}_F(x) \wedge q - r \in \mathbb{Q}_F(y)) \\ &\iff \exists r \in \mathbb{Q}\ (r \in \mathbb{Q}_L(\varphi(x)) \wedge q - r \in \mathbb{Q}_L(\varphi(y))) \\ &\iff q \in \mathbb{Q}_L(\varphi(x) + \varphi(y)), \end{aligned}$$

by $(\star)$, Lemma 6.2.1 in F, $(\star)$ twice, and Lemma 6.2.1 in L. By Lemma 6.1.3 in L, $\varphi(x+y) = \varphi(x) + \varphi(y)$.

Step 6: products and the unit. First let $x > 0$ and $y > 0$. By Steps 3 and 4, $\varphi(x) > \varphi(0_F) = 0_L$ and $\varphi(y) > 0_L$; and $xy > 0$, $\varphi(x)\varphi(y) > 0$ by (O4). Let $q \in \mathbb{Q}$. If $q \leq 0$, then $q \in \mathbb{Q}_F(xy) = \mathbb{Q}_L(\varphi(xy))$, and $q \in \mathbb{Q}_L(\varphi(x)\varphi(y))$, by the last clause of Lemma 6.2.2 in F and in L. If $q > 0$, then

$$\begin{aligned} q \in \mathbb{Q}_L(\varphi(xy)) &\iff q \in \mathbb{Q}_F(xy) \\ &\iff \exists r \in \mathbb{Q}_{>0}\ (r \in \mathbb{Q}_F(x) \wedge q/r \in \mathbb{Q}_F(y)) \\ &\iff \exists r \in \mathbb{Q}_{>0}\ (r \in \mathbb{Q}_L(\varphi(x)) \wedge q/r \in \mathbb{Q}_L(\varphi(y))) \\ &\iff q \in \mathbb{Q}_L(\varphi(x)\varphi(y)), \end{aligned}$$

by $(\star)$, Lemma 6.2.2 in F, $(\star)$ twice, and Lemma 6.2.2 in L (with $\varphi(x), \varphi(y) > 0$). By Lemma 6.1.3 in L, $\varphi(xy) = \varphi(x)\varphi(y)$ for $x, y > 0$.

Now the signs. By Steps 5 and 3, $\varphi(x) + \varphi(-x) = \varphi(0_F) = 0_L$, so $\varphi(-x) = -\varphi(x)$ (Proposition 2.3.2 (a)). If $x = 0$ or $y = 0$, both sides of $\varphi(xy) = \varphi(x)\varphi(y)$ are 0_L (Step 3 and Proposition 2.3.2 (b)). If $x < 0 < y$, then $-x > 0$ (Proposition 2.3.5 (a)) and $xy = -((-x)y)$ (Proposition 2.3.2 (c)), so

$$\varphi(xy) = -\varphi((-x)y) = -(\varphi(-x)\varphi(y)) = -(-\varphi(x))\varphi(y) = \varphi(x)\varphi(y).$$

The case $x > 0 > y$ follows by (F6), and if $x, y < 0$, then $xy = (-x)(-y)$ and $\varphi(xy) = (-\varphi(x))(-\varphi(y)) = \varphi(x)\varphi(y)$. Finally $\varphi(1_F) = 1_L$ by Step 3. With Steps 4 and 5, φ is an order-preserving field map.

Step 7: uniqueness. Let $\chi\colon F \to L$ be any order-preserving field map, and let $x \in F$, $q \in \mathbb{Q}$. By Proposition 6.1.1, $\chi(\iota_F(q)) = \iota_L(q)$. If $\iota_F(q) < x$, then $\iota_L(q) = \chi(\iota_F(q)) < \chi(x)$. If $\iota_L(q) < \chi(x)$, that is, $\chi(\iota_F(q)) < \chi(x)$, then $\iota_F(q) < x$, because χ reflects order (the remark after Definition 6.1.1). So

$$\mathbb{Q}_L(\chi(x)) = \mathbb{Q}_F(x) = \mathbb{Q}_L(\varphi(x)),$$

the second equality by $(\star)$. By Lemma 6.1.3 in L, $\chi(x) = \varphi(x)$. Since x was arbitrary, $\chi = \varphi$. This proves the theorem.

■

The answer to the prompt. Completeness of L is used in Step 1, where the supremum must exist, and in Step 2 ($\supseteq$), where its leastness turns "below $\varphi(x)$" into "below some $\iota_L(r)$". The Archimedean property of F is used in Step 1 for the bounds $-n$ and m, in Steps 2 and 4 through density in F, and in Steps 5 and 6 through the two lemmas applied in F. Completeness of L is used a third time, indirectly: it makes L Archimedean (Theorem 6.1.2), which Steps 3, 5, 6 and 7 need for rationals to separate in L and for the lemmas in L.

Remark. Step 7 is the whole of "uniqueness". Its idea, comparing the rationals below two candidate values, needs only that L is Archimedean, not that it is complete: Exercise C1 runs it for any two order-preserving field maps $F \to L$, with no appeal to $(\star)$. Section 6.3 leans on the uniqueness clause throughout: to show that the maps between two complete fields are inverse to each other, that the isomorphism they give is the only one, and that a complete ordered field has no order-preserving field automorphism other than the identity.

Remark. Neither hypothesis can be dropped from the existence half. *Without completeness of L:* take $F = \mathbb{R}$, which is Archimedean (Theorem 5.4.2), and $L = \mathbb{Q}$. An order-preserving field map $\varphi\colon \mathbb{R} \to \mathbb{Q}$ would fix 2 (Proposition 6.1.1), so for $\sqrt{2} \in \mathbb{R}$ (the element of Example (The Class of the Newton Sequence), as in Section 6.1), $\varphi(\sqrt{2})^2 = \varphi((\sqrt{2})^2) = \varphi(2) = 2$ in $\mathbb{Q}$, against Theorem 2.6.1. *Without the Archimedean property of F:* take $F = \mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field) and any complete L. An order-preserving field map φ would fix each $n \cdot 1$ and be strictly increasing, so $n \cdot 1_L = \varphi(n \cdot 1) < \varphi(t)$ for every $n \in \mathbb{N}$, against the Archimedean property of L (Theorem 6.1.2).

Warning. The supremum in the definition of φ is taken **in** L, of the copies $\iota_L(q)$. It cannot be taken in $\mathbb{Q}$, or in F. In $\mathbb{Q}$: the set $\mathbb{Q}_{\mathbb{R}}(\sqrt{2})$ of Example (The Rationals below the Square Root of Two) (with $\sqrt{2}$ as in Section 6.1) has no supremum in $\mathbb{Q}$. It contains the set $A = \{p \in \mathbb{Q} : p^2 < 2\}$ of Theorem 2.6.2, and its other elements are negative, so below $0 \in A$; hence the two sets have the same upper bounds in $\mathbb{Q}$, and A has no least one. In F: the supremum of $\iota_F(\mathbb{Q}_F(x))$ always exists and is x itself, since x is an upper bound and, by density in F (Theorem 4.2.5), every $u < x$ lies below some $\iota_F(q) < x$. So it is an element of the wrong field.

Quick check. Take $L := F$, a complete ordered field. Which map $F \to F$ does Theorem 6.2.3 produce, and why, without computing a single supremum?

Solution. The identity. The theorem says there is exactly one order-preserving field map $F \to F$ (F is Archimedean by Theorem 6.1.2), and id_F is one (the first example after Definition 6.1.1). So $\varphi = \mathrm{id}_F$. This is the argument Section 6.3 builds on.

6.2.4 Exercises

6.2.4.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Theorem 6.2.3.

(a) Which steps use that L is complete, and which that F is Archimedean? Which steps use that L is Archimedean, and where does that come from?

(b) Run the definition of φ with $F = \mathbb{Q}$ and $L = \mathbb{R}$, the constructed reals. What is φ?

(c) State the idea of the proof in two sentences.

(d) Why may Step 1 not define $\varphi(x)$ as a supremum taken in F?

Solution.

(a) L complete: Step 1 (the supremum exists) and Step 2 ($\supseteq$) (its leastness). F Archimedean: Step 1 (the bounds $-n$, m), Steps 2 and 4 (density in F), and Steps 5 and 6 (Lemma 6.2.1 and Lemma 6.2.2 applied in F). L Archimedean: Steps 3, 5, 6 and 7, through Lemma 6.1.3 in L and the two lemmas in L. It comes from the completeness of L, by Theorem 6.1.2.

(b) $\varphi = \iota_{\mathbb{R}} = \kappa$. By Step 3, $\varphi(\iota_{\mathbb{Q}}(r)) = \iota_{\mathbb{R}}(r)$ for every $r \in \mathbb{Q}$, and $\iota_{\mathbb{Q}}$ is the identity (Chapter 4, after Theorem 4.2.2). So $\varphi(r) = \iota_{\mathbb{R}}(r)$, which is $\kappa(r)$ by Theorem 5.4.1 (d). Concretely, $\varphi(r)$ is the supremum in $\mathbb{R}$ of $\{\kappa(q) : q < r\}$, and that supremum is $\kappa(r)$.

(c) An order-preserving map that fixes the rationals must send x to an element with the same rationals below it, and in a complete L the supremum of their copies is such an element. Every field identity for φ is then checked by comparing the rationals below the two sides, which the sum and product lemmas compute, and rationals separate in L.

(d) In F the supremum of $\iota_F(\mathbb{Q}_F(x))$ always exists, since F is Archimedean, and it is x itself (Exercise B2 of Section 6.1): an element of F, not of L. So it does not give a map into L. The supremum must be of the copies $\iota_L(q)$, taken in L.

6.2.4.2 B. Practice

Exercise. B1

Prove Lemma 6.2.2: let F be an Archimedean ordered field, $a, b \in F$ with $a, b > 0$, and

$q \in \mathbb{Q}$ with $q > 0$; show that $\iota_F(q) < ab$ if and only if some rational $r > 0$ has $\iota_F(r) < a$ and $\iota_F(q/r) < b$. Show also that $\iota_F(q) < ab$ for every rational $q \leq 0$.

Hint: imitate the proof of Lemma 6.2.1, with division in place of subtraction.

Solution. Write q for $\iota_F(q)$. Since ι_F preserves products, $\iota_F(r)\,\iota_F(r^{-1}) = \iota_F(1) = 1$ for $r \neq 0$, so $\iota_F(r^{-1}) = \iota_F(r)^{-1}$ (Lemma 4.1.1 (a)) and $\iota_F(q/r) = q/r$ in F.

If $q \leq 0$, then $q \leq 0 < ab$ by (O4).

Let $q > 0$. ($\Leftarrow$) Let $r > 0$, $r < a$, $q/r < b$. Then $q/r > 0$ (Proposition 2.5.2 (a), (O4)), so Proposition 2.3.5 (c) gives $q = r(q/r) < a(q/r)$, and, as $a > 0$, $a(q/r) < ab$. Hence $q < ab$.

($\Rightarrow$) Let $q < ab$. Since $b^{-1} > 0$, Proposition 2.3.5 (c) gives $q/b < a$, and $q/b > 0$. By Theorem 4.2.5 there is $r \in \mathbb{Q}$ with $q/b < \iota_F(r) < a$. Then $\iota_F(r) > 0 = \iota_F(0)$, so $r > 0$ since ι_F reflects order. From $q/b < r$ and $b > 0$, $q < rb$; multiplying by $r^{-1} > 0$, $q/r < b$. This r is as required.

Exercise. B2: Cube roots

Let F be a complete ordered field and $a \in F$ with $a > 0$. Let $S := \{y \in F : y \geq 0 \text{ and } y^3 < a\}$. The following steps prove that some $s \in F$ has $s^3 = a$.

(1) S is non-empty ($0 \in S$) and bounded above by $1 + a$: if $y > 1 + a$, then $y > 1$, so $y^3 > y > a$. Let $s := \sup S$. Then $s > 0$, since $m := \min\{1, a/2\}$ has $0 < m \leq 1$, so $m^3 \leq m \leq a/2 < a$, and $m \in S$.

(2) For $y, z \in F$, $z^3 - y^3 = (z - y)(z^2 + zy + y^2)$ and $z^2 + zy + y^2 = (z + y/2)^2 + (3/4)y^2$, which is > 0 unless $y = z = 0$ (if $y \neq 0$, then $(3/4)y^2 > 0$ by Proposition 2.3.5 (e), (c) and the square is ≥ 0, so the sum is positive by (b); if $y = 0$ and $z \neq 0$, the expression is $z^2 > 0$ by (e)). Hence $y < z \iff y^3 < z^3$, for all $y, z \in F$.

(3) The case $s^3 < a$ is impossible. Let $h := \min\{1, (a - s^3)/(2(3s^2 + 3s + 1))\} > 0$. Since $0 < h \leq 1$ and $s \geq 0$, $(s + h)^3 = s^3 + 3s^2h + 3sh^2 + h^3 \leq s^3 + h(3s^2 + 3s + 1) \leq s^3 + (a - s^3)/2 < a$, so $s + h \in S$, against s being an upper bound.

(4) The case $s^3 > a$ is impossible. *Name the move, then complete it.*

(5) Hence $s^3 = a$.

Then explain how to get a cube root of $a = 0$ and of $a < 0$, and why every $a \in F$ has **exactly one** cube root.

Solution. The move is **construct a point as a supremum**: s is built as $\sup S$, and each of the two wrong cases of trichotomy is refuted by exhibiting a point that contradicts one clause of the supremum. In step (3) it was an element of S above s, against (S1). In step (4) it is an upper bound below s, against (S2).

(4) Suppose $s^3 > a$. Let $h := (s^3 - a)/(3s^2)$; it is positive, since $s > 0$ by (1). Also $h < s$: this says $s^3 - a < 3s^3$, which holds because $a > 0$ and $2s^3 > 0$. So $s - h > 0$, and

$$(s - h)^3 - a = (s^3 - a) - 3s^2h + 3sh^2 - h^3 = h^2(3s - h) > 0,$$

since $3s^2h = s^3 - a$, $h^2 > 0$ and $3s - h > 2s > 0$. Let $y > s - h$. By (2), $y^3 > (s - h)^3 > a$,

so $y \notin S$. Hence every element of S is $\leq s-h$, and $s-h$ is an upper bound of S smaller than s, against s being the least upper bound.

(5) Steps (3) and (4) leave only $s^3 = a$, by trichotomy (O1).

For $a = 0$, $0^3 = 0$. For $a < 0$, $-a > 0$ has a cube root c by the above, and $(-c)^3 = -c^3 = a$ (Proposition 2.3.2 (c)). Uniqueness: if $b^3 = c^3$, then neither $b < c$ nor $c < b$ can hold, since by (2) either would make the cubes differ; so $b = c$. So every element of a complete ordered field has exactly one cube root.

6.2.4.3 C. Going deeper

Exercise. C1: Exit check: at most one map into an Archimedean field

Let F and L be Archimedean ordered fields, and let $\chi_1, \chi_2 \colon F \to L$ be order-preserving field maps. Prove that $\chi_1 = \chi_2$.

Solution (Hint 1). Compare the rationals below $\chi_1(x)$ and below $\chi_2(x)$.

Solution (Hint 2). By Proposition 6.1.1, $\chi_i(\iota_F(q)) = \iota_L(q)$ for $i = 1, 2$ and every $q \in \mathbb{Q}$.

Solution. Let $x \in F$ and $i \in \{1, 2\}$. By Proposition 6.1.1 (which needs no order hypothesis), $\chi_i(\iota_F(q)) = \iota_L(q)$ for every $q \in \mathbb{Q}$. Since χ_i is strictly increasing and reflects order (the remark after Definition 6.1.1),

$$\iota_F(q) < x \iff \chi_i(\iota_F(q)) < \chi_i(x) \iff \iota_L(q) < \chi_i(x).$$

So $\mathbb{Q}_L(\chi_1(x)) = \mathbb{Q}_F(x) = \mathbb{Q}_L(\chi_2(x))$. Since L is Archimedean, Lemma 6.1.3 gives $\chi_1(x) = \chi_2(x)$. As x was arbitrary, $\chi_1 = \chi_2$. (This is Step 7 of Theorem 6.2.3 with χ_2 in place of φ; completeness of L is not needed, and neither is the Archimedean property of F.)

Why L must be Archimedean (remark, not proved here). The last step separates $\chi_1(x)$ from $\chi_2(x)$ by a rational strictly between them, which is density of $\iota_L(\mathbb{Q})$ in L, and density needs Archimedes (Theorem 4.2.5). In the non-Archimedean field $\mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field), the elements $1/t$ and $2/t$ are different and have the same rationals below them, namely all $q \leq 0$: a positive rational q has $q > 1/n$ in $\mathbb{Q}$ for some $n \in \mathbb{Z}_{>0}$ (Theorem 2.5.3), so $\iota_{\mathbb{Q}(t)}(q) > \iota_{\mathbb{Q}(t)}(1/n) = 1/(n \cdot 1) > 2/t$, because $t > 2n \cdot 1$ (Example (A Non-Archimedean Ordered Field)). So the separating step fails in such a field. For the stronger statement proved above, with F arbitrary, the hypothesis on L cannot be dropped: on $\mathbb{Q}(t)$, the substitution $t \mapsto t + 1$ keeps degrees and leading coefficients, so it is an order-preserving field map $\mathbb{Q}(t) \to \mathbb{Q}(t)$ different from the identity (school polynomial algebra, as flagged in Example (A Non-Archimedean Ordered Field)). An example with F Archimedean needs more algebra than this volume has, and we give none.

Marking scheme. (10 points) - 2: each χ_i fixes $\mathbb{Q}$, with the reason (Proposition 6.1.1, or directly: induction for $\mathbb{N}$, then negatives and quotients). - 3: links the rationals below x to those below $\chi_i(x)$, using that χ_i fixes $\mathbb{Q}$ and reflects order. Either gives the full equivalence $\iota_F(q) < x \iff \iota_L(q) < \chi_i(x)$, or, for the squeeze route of the next item, applies reflection

by χ_1 and χ_2 to a rational strictly between $\chi_1(x)$ and $\chi_2(x)$. - 3: concludes $\chi_1(x) = \chi_2(x)$ by rationals separating, or by a rational strictly between the two, invoking the Archimedean property **of** L. - 2: states the conclusion for every $x \in F$. Fatal errors (0 for the step): using density in L without a hypothesis that gives it; assuming χ_i surjective.

Exercise. C2: Dense image, and when the map is onto

Let F be an Archimedean ordered field, L a complete ordered field, and $\varphi\colon F \to L$ the map of Theorem 6.2.3.

(a) Prove that $\varphi(F)$ is dense in L: for all $u < v$ in L there is $x \in F$ with $u < \varphi(x) < v$.

(b) Prove that if F is complete, then φ is surjective.

(c) Prove that if φ is surjective, then F is complete.

Solution (Hint 1). For (b), given $y \in L$, look at the set $S \coloneqq \mathbb{Q}_L(y)$ and its copy $\iota_F(S)$ in F.

Solution (Hint 2). For (b): $\iota_F(S)$ is non-empty and bounded above in F; let $x \coloneqq \sup_F \iota_F(S)$, and show $\mathbb{Q}_F(x) = S$.

Solution. By Theorem 6.1.2, L is Archimedean.

(a) Let $u < v$ in L. By Theorem 4.2.5 in L there is $q \in \mathbb{Q}$ with $u < \iota_L(q) < v$. By Step 3 of the proof of Theorem 6.2.3, $\iota_L(q) = \varphi(\iota_F(q))$, so $x \coloneqq \iota_F(q)$ works.

(b) Let F be complete and $y \in L$. Let $S \coloneqq \mathbb{Q}_L(y)$. Since L is Archimedean, there are $n, m \in \mathbb{N}$ with $-y < \iota_L(n)$ and $y < \iota_L(m)$. So $-n \in S$, and every $q \in S$ has $\iota_L(q) < y < \iota_L(m)$, hence $q < m$ and $\iota_F(q) < \iota_F(m)$. So $\iota_F(S)$ is non-empty and bounded above in F, and $x \coloneqq \sup_F \iota_F(S)$ exists.
We show $\mathbb{Q}_F(x) = S$. ($\supseteq$) Let $q \in S$. By density in L there is $r \in \mathbb{Q}$ with $\iota_L(q) < \iota_L(r) < y$; then $r \in S$ and $q < r$, so $\iota_F(q) < \iota_F(r) \le x$. ($\subseteq$) Let $\iota_F(q) < x$. Then $\iota_F(q)$ is not an upper bound of $\iota_F(S)$, so $\iota_F(q) < \iota_F(r)$ for some $r \in S$; hence $q < r$ and $\iota_L(q) < \iota_L(r) < y$, so $q \in S$.
By ($\star$), $\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x) = S = \mathbb{Q}_L(y)$, so $\varphi(x) = y$ by Lemma 6.1.3 in L.

(c) Let φ be surjective. It is injective (strictly increasing), so bijective, with an inverse $\varphi^{-1}\colon L \to F$ (Sets and Functions, Theorem 2.4.4). Let $A \subseteq F$ be non-empty with an upper bound b. Then $\varphi(A)$ is non-empty and bounded above by $\varphi(b)$, since $a \le b$ gives $\varphi(a) \le \varphi(b)$. As L is complete, $s \coloneqq \sup \varphi(A)$ exists; let $c \coloneqq \varphi^{-1}(s)$. *Upper bound:* for $a \in A$, $\varphi(a) \le s = \varphi(c)$, so $a \le c$, because φ reflects order (and is injective, for the case of equality). *Least:* if u is an upper bound of A, then $\varphi(u)$ is an upper bound of $\varphi(A)$, so $\varphi(c) = s \le \varphi(u)$, and $c \le u$ as before. So $c = \sup A$ (Sets and Functions, Definition 5.3.5), and F is complete.

So φ is an order-preserving field isomorphism exactly when F is complete. Section 6.3 reaches the "if" direction by a different route, which computes no supremum at all.

Marking scheme. (10 points) - 2: (a) L is Archimedean (Theorem 6.1.2), so some $\iota_L(q)$ lies strictly between u and v, and $\iota_L(q) = \varphi(\iota_F(q))$ (Step 3). - 2: (b) $\iota_F(S)$ is non-empty and

bounded above in F (Archimedes in L, transferred through ι_F), so $x = \sup_F \iota_F(S)$ exists by completeness of F. - 2: (b) $\mathbb{Q}_F(x) = S$, both inclusions; ($\supseteq$) uses density in L. - 1: (b) $\mathbb{Q}_L(\varphi(x)) = \mathbb{Q}_F(x) = \mathbb{Q}_L(y)$ by ($\star$), hence $\varphi(x) = y$ by Lemma 6.1.3 in L. - 3: (c) for non-empty $A \subseteq F$ bounded above by b, $\varphi(A)$ is bounded above by $\varphi(b)$, and $\varphi^{-1}(\sup \varphi(A))$ is an upper bound of A and the least one, both using that φ reflects order and is bijective. Fatal errors (0 for the step): using density of $\iota_L(\mathbb{Q})$ in L without the Archimedean property of L; in (b), taking a supremum in F before checking that the set is non-empty and bounded above; in (c), assuming φ^{-1} preserves order without justification.